Self-Supervised Diffusion for Topology Robustness Optimization in Internet of Things

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***Abstract*—With the rapid expansion of the Internet of Things (IoT), large-scale networks have become increasingly vulnerable to node failures, making robust topology design essential for maintaining reliable connectivity and Quality of Service (QoS). However, existing edge-swap–based optimization methods inherently suffer from path dependence, often converging to structurally biased and suboptimal topologies. To address this limitation, we propose a diffusion-based self-supervised framework that combines a spatio-temporal embedding–driven diffusion strategy, which learns robustness-related structural priors from unlabeled topologies, with a constraint-aware edge-attribute decoding strategy that incorporates domain constraints to generate geometrically feasible, connected, and highly robust networks. Experiments across varying densities and scales demonstrate state-of-the-art robustness improvements, fast convergence, and competitive runtime compared with RL baselines, underscoring the practicality of the proposed approach for IoT systems.**

***Index Terms*—Internet of Things, network topology, robustness optimization, diffusion model**

I. Introduction

Recently, the Internet of Things (IoT) has become integral to critical domains such as industrial production [1], intelligent transportation [2], healthcare [3], and smart homes [4]. As IoT deployments scale up, the interconnection of large numbers of low-power devices and sensor nodes gives rise to increasingly complex network structures, where reliability and connectivity are paramount. However, node failures caused by hardware malfunctions or cyberattacks can disrupt network traffic and even trigger cascading failures, ultimately degrading the overall Quality of Service (QoS) [5]. Therefore, enhancing topology robustness [6] (formally defined in Sec. II) is essential to ensure sustained QoS in large-scale IoT systems.

Existing search-based methods, such as the hill-climbing algorithm (HCA) [7] and the evolutionary framework ROCKS [8], optimize IoT topologies through iterative local edge swaps under connectivity constraints. Reinforcement learning–based variants (e.g., DDLP [9] and ROMEM [10]) accelerate this process but still operate on local modifications. However, these edge-wise operations inherently introduce path dependence, where early edge selections constrain subsequent choices and progressively narrow the search space [11]. As suboptimal decisions accumulate, many potentially robust configurations become inaccessible—a structural limitation that cannot be eliminated through heuristic refinements. Consequently, such methods often converge to biased configurations and struggle to discover genuinely high-robustness topologies.

Data-driven techniques—such as backpropagation-based predictors [12], GNN architectures [13], and diffusion-style predictors [14]—offer theoretical potential to alleviate path dependence by exploring topological distributions in latent spaces and enabling global, parallel optimization. However, when applied to topology robustness optimization, these approaches still lack practical effectiveness due to two fundamental challenges: (*i*) reliance on large labeled datasets, which are often unavailable in IoT scenarios; and (*ii*) the absence of decoding mechanisms that incorporate IoT-specific constraints [15], frequently resulting in infeasible or suboptimal topologies.

To address these challenges, this paper proposes a unified diffusion-based self-supervised framework for IoT topology robustness optimization, as illustrated in Fig. 1. For challenge (*i*), we introduce a spatio-temporal topology embedding with a noise-injection diffusion strategy, which encodes node positions, edge connectivity, and diffusion timesteps into a continuous latent space. Within this space, controlled noise enables self-supervised learning of robustness-related structural priors directly from unlabeled topologies, thereby reducing label dependence and mitigating path dependence. For challenge (*ii*), we develop a constraint-aware edge-attribute decoding strategy that integrates communication range, degree balance, and component-bridging constraints to transform the diffusion-generated confidence matrix into geometrically feasible, connected, and robust network topologies. Through iterative denoising, these two strategies jointly refine topology embeddings into fault-tolerant configurations, achieving global optimization beyond edge-level heuristics. Extensive experiments across diverse network settings verify the framework's superior robustness and competitive runtime compared with evolutionary and RL-based baselines.

The main contributions of this work are summarized as follows:

- We propose a diffusion-based self-supervised framework

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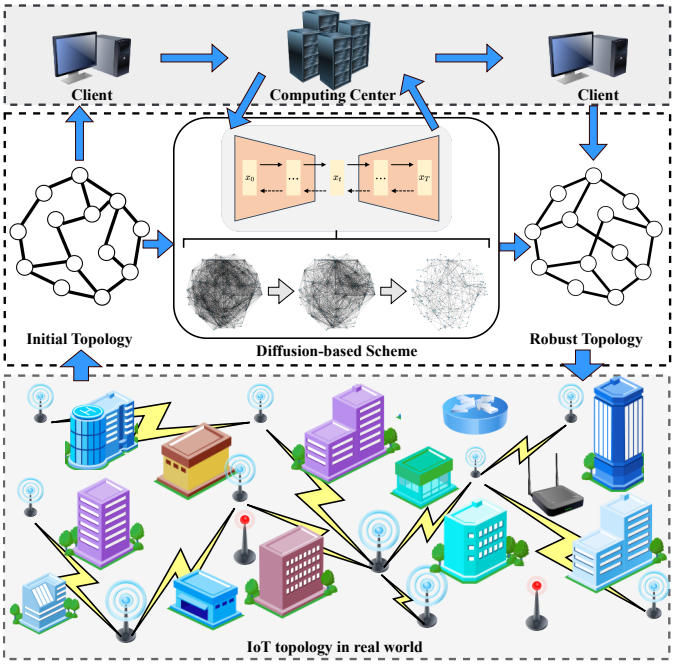


Fig. 1. The proposed scheme for IoT topology robustness.

that incorporates a spatio-temporal embedding and noise-injection diffusion strategy to learn robustness-related structural priors from unlabeled topologies, reducing label dependence and mitigating path dependence.

- We develop a constraint-aware edge-attribute decoding strategy that ensures the geometric feasibility, connectivity, and robustness of reconstructed topologies.
- Extensive experiments across diverse network settings verify the framework's superior robustness and competitive runtime over evolutionary and RL-based baselines.

The remainder of this paper is organized as follows: Section II formulates the problem statement, Section III presents the proposed diffusion-based methodology, Section IV evaluates performance against benchmarks, and Section V concludes the paper with future directions.

II. Problem Statement

A. Topology Robustness

We evaluate network robustness using the High-Degree Attack (HDA) strategy [16], which simulates sequential removals of the highest-degree nodes. After each removal, the size of the maximum connected subgraph, $MCS(i)$, is recorded. The robustness index R is defined as:

$$R = \frac{1}{N+1} \sum_{i=1}^{N} \frac{MCS(i)}{N}. \tag{1}$$

where N is the number of nodes. For a fully connected topology, $R = 0.5$, implying that $R \in (0, 0.5]$. A higher R indicates stronger resilience against cascading node failures.

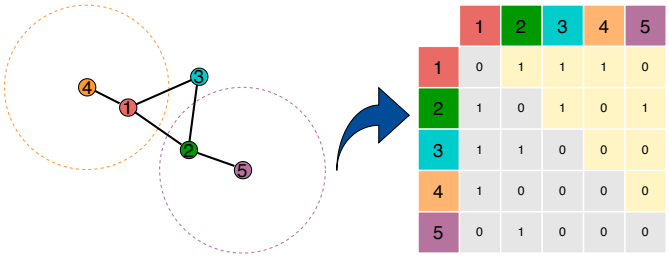


Fig. 2. Illustrative IoT topology (left) and its symmetric adjacency matrix A (right). Nodes are fixed after deployment; dashed circles mark per-node communication range D. An entry $A_{ij} = 1$ denotes a bidirectional link; $A_{ij} = 0$ denotes no link.

B. IoT Topology Modeling and Optimization

We represent an IoT network as an undirected simple graph $G = (V, E)$ embedded in a 2D plane, where each node $v_i \in V$ has a position $\text{pos}(v_i) \in S \subset \mathbb{R}^2$. Two nodes are *physically feasible* to connect if their Euclidean distance satisfies $\text{dis}(v_i, v_j) \leq D$, where D is the maximum communication range. Edges violating this constraint ($\text{dis} > D$) are considered *illegal* and excluded from the feasible edge set $\mathcal{E}_{\text{feas}}$.

The IoT topology considered in this work is static, meaning that all node positions are fixed once deployed and remain unchanged during optimization. Robustness is improved purely through virtual rewiring by reselecting a subset of feasible communication links from $\mathcal{E}_{\text{feas}}$ to form a new edge configuration that enhances overall connectivity and fault tolerance, without physically moving any node. The network connectivity is represented by a symmetric binary adjacency matrix $A \in \{0,1\}^{N \times N}$, where $A_{ij} = 1$ if $(v_i, v_j) \in E$ and $A_{ii} = 0$. An illustrative example is shown in Fig. 2.

To ensure fair comparison, the number of active edges is kept constant ($|E_{\text{before}}| = |E_{\text{after}}|$), i.e., optimization only rearranges feasible links without changing node or edge counts. The key notations used in this section are summarized in Table I.

TABLE I
Notation for IoT topology and geometric constraints

Symbol	Description
$G = (V, E)$	IoT topology
N	Number of nodes
M	Edge density (average degree)
A	Symmetric adjacency matrix
S	Deployment region
R	Topology robustness
D	Communication range
$\text{pos}(v_i)$	Spatial coordinate of node v_i
$\text{dis}(v_i, v_j)$	Euclidean distance between v_i and v_j
$\mathcal{E}_{\text{feas}}$	Feasible edge set ($\text{dis} \leq D$)

Formally, given an initial topology $G_0 = (V, E_0)$ and the feasible edge set $\mathcal{E}_{\text{feas}} = \{(i, j) : \text{dis}(v_i, v_j) \leq D\}$, the

topology robustness optimization can be formulated as:

$$\begin{aligned}
&\textbf{Given: } G_0 = (V, E_0),\ \mathcal{E}_{\text{feas}} = \{(i,j) : \text{dis}(v_i, v_j) \le D\}.\\
&\textbf{Find: } G^* = (V, E^*),\ E^* \subseteq \mathcal{E}_{\text{feas}}.\\
&\textbf{Max: } R(G^*) = \frac{1}{N+1}\sum_{i=1}^{N}\frac{MCS(i)}{N}.\\
&\textbf{s.t. } |E^*| = |E_0|,\ \text{pos}(v_i) \in S,\ \forall i,\ A_{ij} = A_{ji} \in \{0,1\}.
\end{aligned} \tag{2}$$

This formulation ensures the same number of active edges before and after rewiring and restricts the optimization to feasible edges, ensuring geometric validity and node immobility.

III. Methodology

We propose a diffusion-based framework for IoT topology robustness optimization, as illustrated in Fig. 3. The framework consists of two key stages: (Step 1) spatio-temporal topology embedding and self-supervised diffusion modeling, which learn robustness-related structural priors through controlled noise injection in a latent space; and (Step 2) constraint-aware edge-attribute decoding, which reconstructs physically feasible and highly robust topologies from the diffusion outputs.

A. Spatio-Temporal Topology Embedding

In Step 1, each IoT topology is first encoded into spatio-temporal embeddings for nodes, edges, and diffusion timesteps. Given node coordinates x_i, adjacency entry A_{ij}, and diffusion timestep t, the embeddings are defined as:

$$\begin{cases}
\mathbf{h}_i^0 = [\sin(\omega x_i),\, \cos(\omega x_i)]\mathbf{W}_n,\\
\mathbf{e}_{ij}^0 = [\sin(\omega A_{ij}),\, \cos(\omega A_{ij})]\mathbf{W}_e,\\
\mathbf{z}_t = [\sin(\omega t),\, \cos(\omega t)]\mathbf{W}_t.
\end{cases} \tag{3}$$

where $\sin(\cdot)$ and $\cos(\cdot)$ act elementwise on coordinates and adjacency entries; ω denotes a learnable frequency vector, and $\mathbf{W}_n$, $\mathbf{W}_e$, and $\mathbf{W}_t$ are projection matrices. These embeddings jointly preserve spatial structure and temporal (timestep) information and serve as inputs to the diffusion backbone.

B. Self-supervised Diffusion Modeling

The Self-supervised diffusion strategy adopts a variational inference framework [17] to model topological edge distributions as latent variables that evolve through complementary *forward* and *reverse* processes [18], [19]. Formally, the marginal likelihood of the observed topology is written as:

$$p_\theta(x_0) = \int p_\theta(x_{0:T})\, dx_{1:T}. \tag{4}$$

where $x_0, \ldots, x_T$ share the same dimensionality. The joint distribution $p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$ captures the reverse process, while the forward process $q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$ sequentially adds noise to the topology embedding. Training minimizes the standard variational bound on the negative log-likelihood:

$$\begin{aligned}
\mathbb{E}[-\log p_\theta(x_0)] &\le \mathbb{E}_q\left[-\log\frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]\\
&= \mathbb{E}_q\left[\sum_{t>1} D_{\text{KL}}(q_t \| p_\theta) - \log p_\theta(x_0|x_1)\right] + C.
\end{aligned} \tag{5}$$

where D_{KL} denotes the Kullback–Leibler divergence, and C collects terms independent of θ.

a) Forward Process: In the continuous diffusion setting, a noise ratio β_t controls the degree of corruption at each step. Concretely, the forward process in diffusion module is defined as: $q(\hat{x}_t|\hat{x}_{t-1}) = \mathcal{N}\Big(\hat{x}_t;\, \sqrt{1-\beta_t}\,\hat{x}_{t-1},\, \beta_t\,\mathbf{I}\Big)$, where $\hat{x}_t$ converges to an isotropic Gaussian as $\prod_{t=1}^{T}(1-\beta_t) \approx 0$. Let $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{\tau=1}^{t}\alpha_\tau$. Then the t-step marginal is defined as:

$$q(\hat{x}_t|\hat{x}_0) = \mathcal{N}\Big(\hat{x}_t;\, \sqrt{\bar{\alpha}_t}\,\hat{x}_0,\, (1-\bar{\alpha}_t)\,\mathbf{I}\Big). \tag{6}$$

Since $\hat{x}_T$ approximates $\mathcal{N}(0, \mathbf{I})$ after sufficient corruption, the forward process ensures broad coverage of possible topological configurations.

b) Reverse Process: During the reverse process, the model aims to reconstruct $\hat{x}_0$ from the noisy sample $\hat{x}_T$. Followed by Bayes' theorem as:

$$q(\hat{x}_{t-1}|\hat{x}_t, \hat{x}_0) = \frac{q(\hat{x}_t|\hat{x}_{t-1}, \hat{x}_0)\, q(\hat{x}_{t-1}|\hat{x}_0)}{q(\hat{x}_t|\hat{x}_0)}. \tag{7}$$

the reverse process remains a closed-form Gaussian distribution, enabling a stepwise denoising guided by a neural network f_θ, so the posterior point estimate $\hat{x}_0$ can be used as:

$$p_\theta(\hat{x}_{t-1}|\hat{x}_t) = q\left(\hat{x}_{t-1}|\hat{x}_t, \hat{x}_0 = \frac{\hat{x}_t - \sqrt{1-\bar{\alpha}_t}\, f_\theta(\hat{x}_t, t)}{\sqrt{\bar{\alpha}_t}}\right). \tag{8}$$

where $\hat{x}_0$ is a point estimate, updated iteratively. After reaching $\hat{x}_0$, we can optionally convert the continuous values into a discrete adjacency matrix or into a probability-like confidence matrix.

c) AGNN Backbone: To capture both node-centric and edge-centric information, we employ an anisotropic graph neural network (AGNN) with edge gating [20] as a backbone of the diffusion model. Unlike conventional GCNs [21] or GATs [22], AGNNs treat edge embeddings as first-class entities, making them especially suited for topology optimization [14]. Denote h_i^ℓ and e_{ij}^ℓ as the node and edge embeddings at layer ℓ with timestep embedding t; the AGNN update is formulated as:

$$\begin{cases}
\hat{\boldsymbol{e}}_{ij}^{\ell+1} = \boldsymbol{A}^\ell \boldsymbol{h}_i^\ell + \boldsymbol{B}^\ell \boldsymbol{h}_j^\ell + \boldsymbol{C}^\ell \boldsymbol{e}_{ij}^\ell,\\
\boldsymbol{e}_{ij}^{\ell+1} = \boldsymbol{e}_{ij}^\ell + \alpha \cdot \mathcal{BN}(\hat{\boldsymbol{e}}_{ij}^{\ell+1}) + \text{MLP}(\mathbf{t}),\\
\boldsymbol{h}_i^{\ell+1} = \boldsymbol{h}_i^\ell + \alpha \cdot \mathcal{BN}\left(\boldsymbol{U}^\ell \boldsymbol{h}_i^\ell + \sum_{j\in\mathcal{N}_i} \sigma(\hat{\boldsymbol{e}}_{ij}^{\ell+1}) \boldsymbol{V}^\ell \boldsymbol{h}_j^\ell\right).
\end{cases} \tag{9}$$

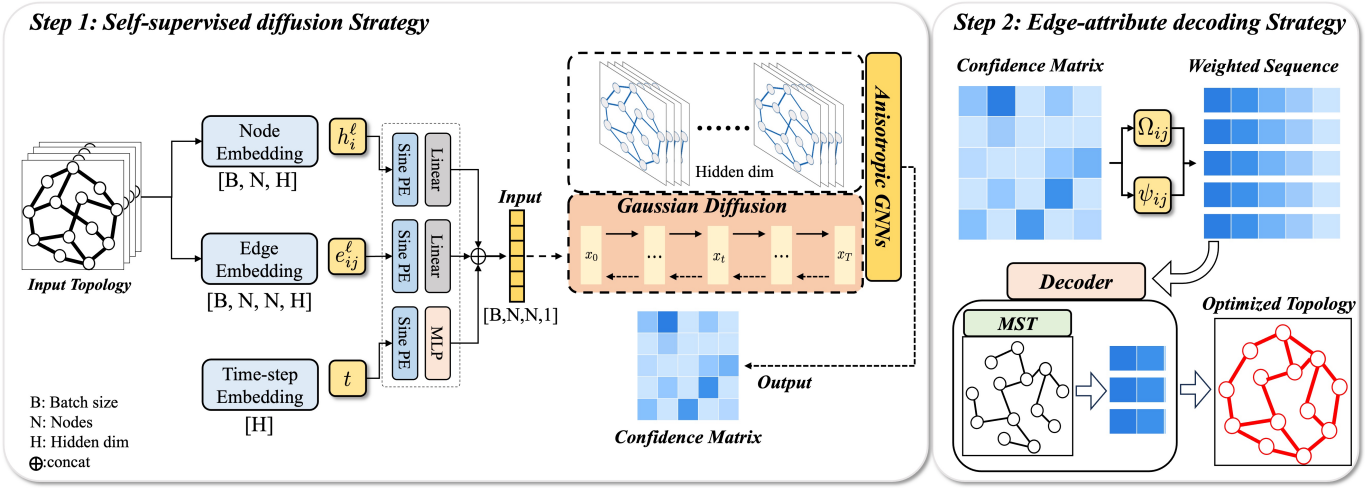


Fig. 3. Overview of the proposed framework.

where $\boldsymbol{A}^{\ell}, \boldsymbol{B}^{\ell}, \boldsymbol{C}^{\ell}, \boldsymbol{U}^{\ell}, \boldsymbol{V}^{\ell}$ are trainable parameters, α is the ReLU activation, $\mathcal{BN}$ denotes batch normalization, σ is the sigmoid function, and $\mathrm{MLP}(\mathbf{t})$ encodes the timestep. In our experiments, we use a 12-layer anisotropic GNN with a width of 256 as described above.

d) Confidence Matrix: Rather than directly sampling discrete edges from a Gaussian, we output a dense confidence matrix whose entries represent probability scores for each edge. This design preserves comparative information on edge likelihood, which is crucial for refining edge selection during the decoding stage [14]. By combining the diffusion mechanism and confidence matrix, our framework can effectively learn and denoise complex topological patterns while avoiding the pitfalls of purely discrete sampling or strict quantization.

C. Edge-attribute decoding Strategy

After obtaining the confidence matrix, we first normalize each entry using a hyperbolic tangent function:

$$A'_{ij} = \frac{\tanh\big(A_{ij}\big) + 1}{2}. \tag{10}$$

which normalizes raw scores into $[0, 1]$ for subsequent processing.

Building on the degree-balance principle [23], we refine A'_{ij} using two operators. First, the *degree-balancing operator* is defined as:

$$\Omega_{ij} = \frac{\exp\Big(-\big(\frac{\Delta d_{ij}}{d_{\max}}\big)^2\Big)}{\sqrt{d_i\, d_j}}, \quad \Delta d_{ij} = \big|\, d_i - d_j \big|. \tag{11}$$

where d_i and d_j are the degrees of nodes i and j, and $d_{\max}$ is the maximum node degree. This operator prevents overemphasis on edges connecting nodes with highly disparate degrees.

Next, the *modified bridging operator* is defined as:

$$\psi_{ij} = \begin{cases} \sqrt{|\mathcal{C}_u| \times |\mathcal{C}_v|}, & \text{if } u \neq v, \\ 1, & \text{if } u = v. \end{cases} \tag{12}$$

where $\mathcal{C}_u$ and $\mathcal{C}_v$ denote the connected components containing nodes u and v prior to adding the edge. This operator assigns higher weights to edges that bridge larger components, thereby promoting global connectivity.

The final score for each edge is computed as:

$$\mathrm{Score}(i, j) = A'_{ij} \times \Omega_{ij} \times \psi_{ij}. \tag{13}$$

Subsequently, edges are sorted in descending order of $\mathrm{Score}(i, j)$. We first construct a minimum spanning tree to guaranty basic connectivity, and then add the remaining edges based on their scores. This strategy avoids the path dependence of iterative methods by selecting edges that best enhance robustness. Moreover, the newly generated topologies can be incorporated into the seed dataset [24], [25] to further strengthen the graph neural network's ability to capture robust topological information.

TABLE II
PARAMETER SETTINGS

Parameter	Value	Comment
S	500 m	Side length
N	$100 - 500$	Number of nodes
D	200 m	Communication range
M	$2 - 6$	Edge density
L	12	Number of GNN layers
H	256	Width of GNN layers
Agg	sum	Aggregation function
lr	0.0002	Learning rate

IV. PERFORMANCE EVALUATION

In this section, we present a comprehensive evaluation of our framework. We detail the parameter settings, assess performance across various network densities and sizes, and compare our framework with baseline algorithms.

A. Parameter Settings

Experiments were conducted under diverse network configurations, as summarized in Table II. We compared our scheme

against several baselines including HCA [7], ROCKS [8], BTW [26], and ROMEM [10]. The BP-NN model [12] was excluded due to its performance not surpassing its training set. Each configuration was subjected to 10 independent trials with randomized initial conditions to ensure statistical robustness. All experiments were implemented in PyTorch and run on an Ubuntu server equipped with a 48-core Intel Xeon Silver 4310 CPU (2.10GHz) and two NVIDIA A40 GPUs.

B. Optimization under Varying Network Densities and Sizes

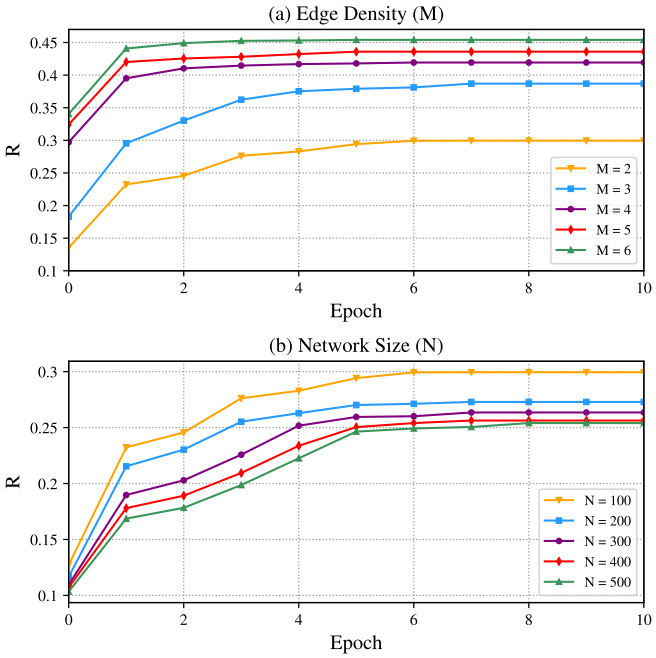


Fig. 4. Optimization performance under varying (a) edge densities and (b) network sizes.

We first evaluated 100-node topologies with edge densities $M \in \{2, 3, 4, 5, 6\}$ and further examined topologies with fixed density $M = 2$ under different network sizes $N \in \{100, 200, 300, 400, 500\}$. Denser networks tend to maintain larger connected subgraphs after node removals, resulting in higher robustness, while larger networks generally exhibit lower robustness due to their increased vulnerability to high-degree attacks.

Fig. 4 illustrates the robustness index R as a function of the number of optimization epochs for both settings. In both cases, our optimization framework rapidly increases R within the first five epochs and subsequently stabilizes, demonstrating fast convergence and stable performance across diverse IoT topologies.

C. Comparison with Baseline Algorithms

We evaluated our scheme with multiple baselines (ROMEM, BTW, ROCKS, HCA) under scenarios in which we varied edge density for a fixed node count and varied node count for a fixed edge density.

a) Varying Edge Density: Fixing the network size at $N = 100$ nodes, we varied the edge density M from 2 to 6. Each setting underwent 10 runs for statistical reliability. Figure 5 demonstrates that higher edge densities generally result in greater robustness R. Notably, BTW exhibits diminished performance beyond $M > 3$, likely due to its local growth-based strategies lacking global topological insight. Table III provides average runtimes for each algorithm in this scenario. Our scheme shows slightly higher execution times than ROMEM but significantly less than other search-based methods, highlighting its balance of robustness and computational efficiency.

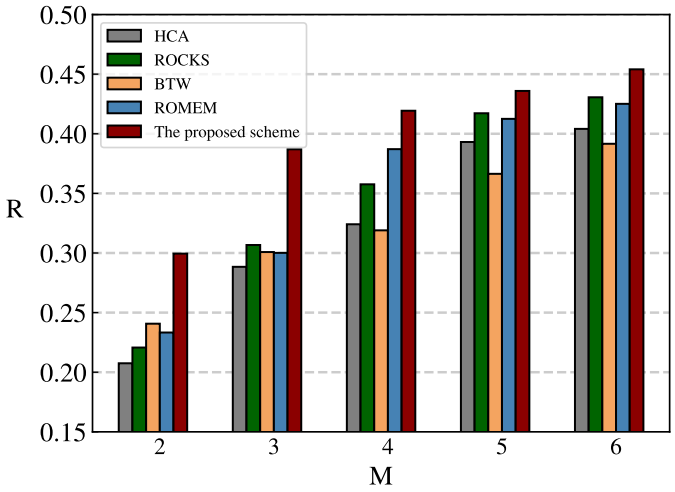


Fig. 5. Robustness with different edge densities.

TABLE III
AVERAGE RUNTIME WITH DIFFERENT EDGE DENSITIES

M	Our Scheme	ROMEM	BTW	ROCKS	HCA
2	7.24E+02	5.44E+02	2.84E+02	9.68E+02	1.69E+03
3	8.41E+02	6.31E+02	5.61E+02	1.06E+03	2.91E+03
4	9.15E+02	6.83E+02	8.98E+02	1.19E+03	4.01E+03
5	9.64E+02	7.14E+02	1.91E+03	1.63E+03	4.72E+03
6	1.34E+03	9.00E+02	2.44E+03	1.81E+03	4.81E+03

b) Varying Network Size: For fixed edge density $M = 2$, we increased node count N from 100 to 500. Figure 6 shows that our scheme consistently achieves higher R values compared to other algorithms as N grows. Regarding runtime (Table IV), our scheme is marginally slower than ROMEM but notably faster than other search-based algorithms, emphasizing its scalability and efficiency for large-scale IoT deployments.

TABLE IV
AVERAGE RUNTIME WITH DIFFERENT NETWORK SIZES

N	Our Scheme	ROMEM	BTW	ROCKS	HCA
100	7.24E+02	5.44E+02	2.84E+02	9.68E+02	1.69E+03
200	6.51E+03	4.31E+03	3.04E+03	4.98E+03	3.25E+04
300	1.31E+04	8.61E+03	1.09E+04	1.74E+04	1.92E+05
400	3.90E+04	2.42E+04	4.45E+04	4.81E+04	7.13E+05
500	5.05E+04	3.08E+04	1.65E+05	1.01E+05	1.85E+06

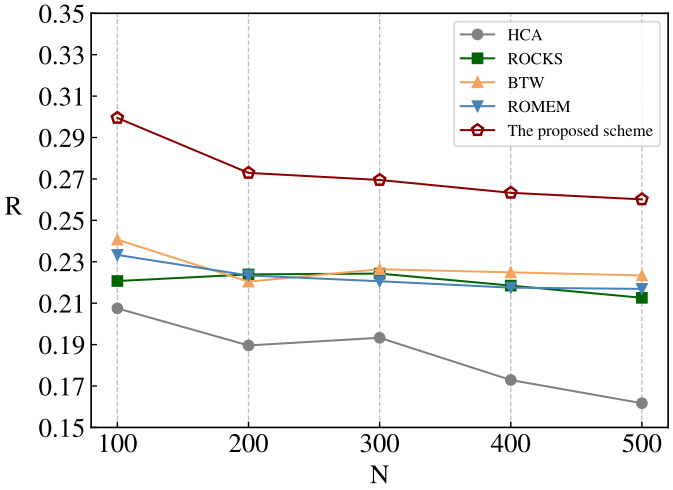


Fig. 6. Robustness with different network sizes.

V. Conclusion

This paper presents a diffusion-based self-supervised framework for IoT topology robustness optimization. By integrating spatio-temporal topology embedding with noise-driven diffusion modeling, the framework learns intrinsic structural priors directly from unlabeled networks, eliminating the reliance on labeled datasets. A constraint-aware edge-attribute decoding strategy further ensures physically feasible, degree-balanced, and globally robust connectivity, effectively overcoming the path dependence inherent in traditional edge-swap–based search methods. Extensive experiments across varying densities and network sizes verify that the proposed approach consistently achieves state-of-the-art robustness improvements, fast convergence, and scalability compared with evolutionary and RL-based baselines. In future work, we plan to extend this framework to dynamic IoT scenarios, explore adaptive diffusion scheduling, and investigate more theoretically grounded decoding mechanisms for enhanced generalizability.

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